

Answer Key — Saturation, Concentration and Solubility

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 (b) **unsaturated** — an unsaturated solution can dissolve more solute.

Q2 (b) **solubility** — this maximum is the solubility.

Q3 (b) **increase** — solubility of most solids increases with temperature.

Q4 (b) **concentrated** — more solute → concentrated solution.

Q5 (a) **True** — heating raises solubility, so it can take more solute.

Q6 **saturated** — that is a saturated solution.

Q7 **concentration** — that is the concentration.

Short answer

Q8 An unsaturated solution can dissolve more solute at that temperature; a saturated solution cannot — extra solute settles at the bottom.

Q9 The maximum amount of a solute that dissolves in a fixed quantity of solvent at a given temperature; for most solids, solubility increases with temperature.

Q10 3 spoons in 100 mL — the same solute is in less solvent, so the solute per mL is higher.

Long / application

Q11 Make a saturated solution of baking soda in water at 20°C (until some stays undissolved). Heat it to 50°C while stirring — the leftover dissolves; add more until saturated again, then heat to 70°C and it dissolves more. This shows hotter water dissolves more solid, so solubility increases with temperature.

Q12 A solution is saturated only at a particular temperature. If you raise the temperature, solubility increases, so it can dissolve more solute and becomes unsaturated. So a saturated solution can dissolve more if heated.
